CHAPTER 2

THE THREE BASICS: LINE, CORNER EXIT SPEED, BRAKING

There are three basic problems to solve in race driving: 1) driving on the best path, 2) carrying speed through corners and onto straights, and 3) efficiently slowing the car at the entry to corners.

LINE FUNDAMENTALS

We know that the fundamental aim of driving through a corner is to drive on an arc with the largest possible radius. Let's put some numbers to the problem, using a simple, straightforward corner like the 90-degree right-hand turn in Fig. 2-1. The dimensions of the corner are a close representation of Turn 7 at the Sebring test circuit. Each of the possible paths has a radius dimension attached to it. The path along the inside of Turn 7 has a radius of 103 feet. The arc around the outside edge of the corner has a radius of 130 feet. The radius of the arc that represents the racing line through the corner is a whopping 195 feet. That's 89% bigger than the inside arc, 50% bigger than going around the outside.

Predicting Corner Speed

To find out how much faster you can go through Turn 7 on the right line, you have to attach some speeds to these arcs. To do this, you have to describe the rela-

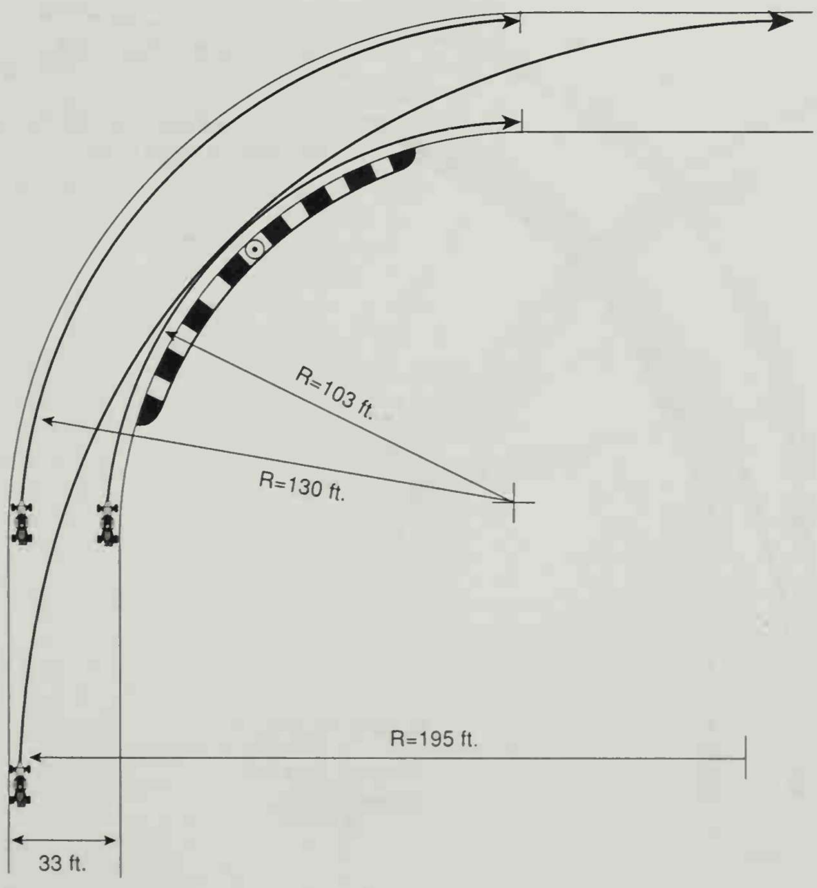


Fig. 2-1. The increase in radius available by driving outside, inside, outside is formidable. In this case the large radius is nearly twice the size of the arc that hugs the inside of the road.

tionship between the size of an arc and the maximum speed a vehicle can attain on that arc. From the skid-pad, you know that the bigger you make the radius, the higher you make the potential speed. In addition, you have to take into account how well the car corners. A car with greater cornering force will reach its limit at a higher speed.

The equation that ties these two interrelated facts together is 15GR = m.p.h.^{2}. G represents the car's maximum cornering force, where G is the force of gravity. R represents the radius of the corner in feet.

To compute the maximum cornering speeds on different arcs through Turn 7, let's simplify things and call our car a 1G car. (For comparison, a street sedan might be able to exert 7/10ths of 1 G of cornering force at maximum. An Indy Car can develop between 1.4 and 4.5 Gs.) Then plug in the radius numbers and solve the equation. For example, to figure out how fast the car could drive around the inside edge of the corner, substitute 1 for G, and 103 feet for R:

$15 \times 1 \times 103 = \text{m.p.h.}^2$

$1545 = \text{m.p.h.}^2$

$39.3 = \text{m.p.h.}$

As you can see in Fig. 2-2, driving around the outside of the corner on the 130-foot radius would allow you to go 44.1 m.p.h. Driving on "the line," with a radius of 195 feet gives you 54 m.p.h., almost 15 m.p.h. faster than hugging the inside of the road, *and* 10 m.p.h. faster than the outside route. You can see why it's a good idea to find the line and to drive on it.

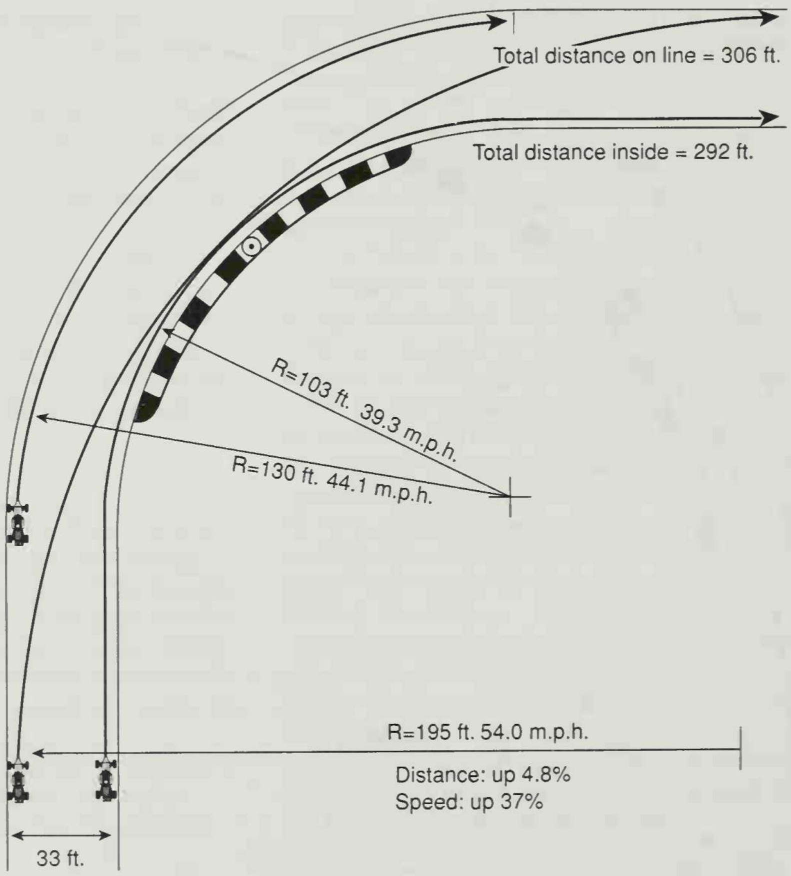
"Nelson Piquet, three-time World Champion, was telling me about him being invited by BMW to do a Pro Car race at the old Nurburgring—14 miles and 176 corners. He'd never driven it before so he said, 'Okay, but I tell you what I want to do—I want three cars and I want to go there for a week.' He took the three cars, blew the motor in one, crashed one, and one he just wore out. But, in the week he did 400 laps of the 14-mile circuit just to learn the racetrack. Nelson won the race, by the way."

—Donny Sullivan

You will also notice that the distance you have to travel on the line is longer than around the inside edge of the corner. Though you would think there'd be an advantage to driving the shorter distance, the tighter arc required serves up a big speed penalty: Although you drive almost 5% further on the line, you do it 37% faster, which is a good trade-off.

Putting the car exactly on the optimum path is not something you strive for because it looks pretty. You

Fig. 2-2. Driving on "the line" results in driving further than taking the shortest path, but in this case, you drive roughly 5% further at 37% greater speed—a good trade-off.



do it because it's fast: it minimizes the time it takes to drive through the corner and maximizes the entry speed onto the following straight.

Turn-in, Apex, and Track-out

For the balance of the discussion in this book, we'll be using specific terms for the reference points of the arc through a corner. It begins at the "turn-in" point, comes closest to the inside edge of the corner at the "apex," and comes closest to the outside edge of the road at the "track-out" point (see Fig. 2-3).

In our example, we've created an arc that starts at the turn-in point, touches the apex and ends at the track-out. You could go out to Turn 7 at Sebring, get to the turn-in point and turn the steering wheel once and hold it there, placing the car on an arc that touches the apex with the right tires, then touches the track-out with the lefts. When you start working on the line, this is exactly what you're looking for, although as you try to get going faster, it's not quite so simple.

The Constant Radius Arc

An arc like this that doesn't change is called a constant radius arc. While you could handle all corners this way—after all, this arc gives you the biggest radius so therefore must allow you the highest speed—it doesn't take into account that in most corners, you are interested in carrying speed *away* from the corner and down the following straight. This requires acceleration, most commonly begun before the apex of a corner. Remember, if you're accelerating in the corner and your speed is rising, the radius of the arc you're on has to increase.

The difficulty is that it is much harder to draw increasing radius arcs than constant radius arcs. An accurate drawing of the arc a racing car is on as it leaves a corner near the limit would be one with an increasing radius. But at the start, you can learn more about identifying the right line by using the simpler version, the constant radius arc. Remember, too, that as you're working on finding the line you'll be driving the car under its limit to start. Only after finding the line—and proving that you can stay on it—will you start adding speed and approach the car's cornering limit.

Reference Points to Keep the Line Consistent

In our racing schools, we place bright orange pylons at the reference points for each corner. The drill looks simple: Get to the turn-in pylon and turn the steering wheel just the right amount to get to the apex pylon—hopefully this arc will end up at the track-out pylon.

The pylons are, at first, a great help in locating reference points, but what each driver eventually needs to do is to find indicators other than pylons around

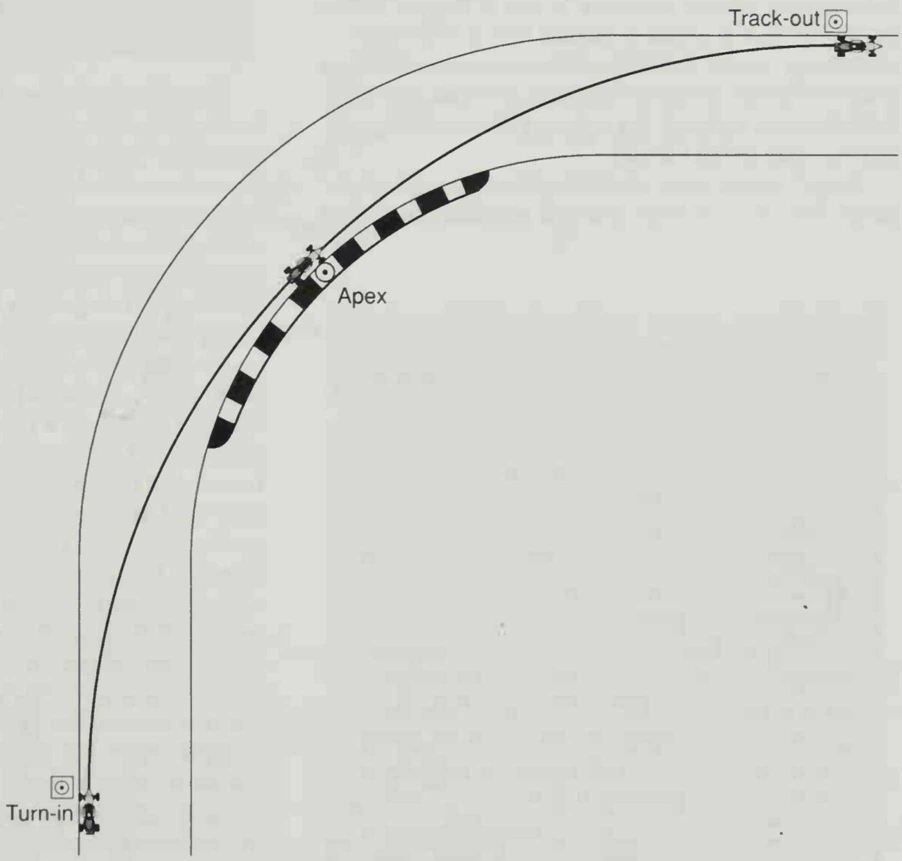


Fig. 2-3. Every corner has at least three distinct reference points: the turn-in, the apex, and the track-out.

"I absolutely use reference points around the track to keep on the line. If they're there, by God, I'll find them. It gives you consistency, and I think that's the difference between consistently running up front and winning races."

—Robbie Buhl

the racetrack that help show them the way. You want reference points that help you consistently turn the car into the corner at the same spot every lap, that help you clip the same apex time after time.

At first, the racetrack may look like a uniform strip of black asphalt. You'll find that you have to look much closer than you might have thought to accurately find your way around. Any kind of irregularity near the turn-in point will do—a gouge in the racing surface, a crack in the pavement, a paint line on the edge of the racetrack, changes in pavement or patches made in the racing surface, curbing at the corner entry.

The further away from the road the reference point is, the more potential for distraction, but, at courses with walls, you can frequently use advertising banners as reference points. Turning in "five feet past the Bosch sign" often works.

It's a mistake to be rigid. The process of finding the line is continuous. If you've never seen the racetrack before, never driven a racecar before, it's going to take a considerable amount of mental energy. You'll go round and round, picking reference points, adjusting them lap by lap until at some point you feel you really have it down, right on the edges of the road, smooth arcs through every corner.

And it's never over. It's difficult to hit marks right on the button every time and you will find that you're constantly fiddling with reference points in the pursuit of the perfect lap. As you'll see in later chapters, there are other variables, beyond your own inability to be perfectly consistent, which will change your corner reference points.

"You have to be willing to experiment. Once you find a line that works you still have to keep looking for something better. Even during a race I'll change lines if I feel I can find a little more grip on another part of the racetrack. I make small changes. I might change my entry to a corner a little bit on one lap and if it's better I might try it a little more the next time. No big jumps. I don't go from the 3 marker to the 1 marker.

"I think that's part of the art of racing—something that all the really good racers do. I think it's one of the things that Al Jr., for example, is really good at. He just keeps plugging and plugging away and keeps working on it and he gets there in the end."

—Bryan Herta

COMMON ERRORS

Inaccuracy

When we set up our pylons in our racing schools, we want to see our drivers get within 6 inches of the turn-in and track-out points; we like to see them touch the apex with their inside tire.

"I was lucky enough when I was a kid to be able to practice all the time. We would go to Thompson in Connecticut in the middle of the week and you could run all day for $5. I just drove and drove and drove—not fast, but I was just logging time, getting experience behind the wheel.

"[We] would go out and put quarters at the apexes, because [we] read somewhere that grand prix drivers could hit a coin at the apex every lap. I got so I could be very precise with the car and when it was time to go fast I was ready."

—David Loring

In racing, inches and tenths of miles per hour matter. For example, if you are one mile per hour faster than your competition, you gain approximately 1.5 feet every second you do this. A 1-m.p.h. advantage in a half hour race represents 2600 feet. At Indianapolis, a race that's close to three hours, a 1-m.p.h. advantage represents a three-mile lead. You find that elusive mile per hour by using every inch of road available. Let's take a look at what happens when you don't.

You've been working on Turn 7 at Sebring, so let's stick with it. If you did it just right, you could drive through Turn 7 on a 195-foot radius at 54 m.p.h. (see earlier Fig. 2-2). If you were 12 inches off the edge of the road at the turn-in, missed the apex by 12 inches, then held the car off the edge of the road at the track-out by 12 inches, your radius would be 189.5 feet. Plug this into our m.p.h. equation and your cornering speed is down to 53.3 m.p.h. Seven-tenths of a mile an hour difference doesn't sound like much, but in terms of the distance lost to a competitor, it's a bunch.

Say you made this mistake at the Keyhole at Mid Ohio, a corner that, in one of our racecars, leads to a straight where you accelerate for 20 seconds. Seven tenths of 1 m.p.h. is still over a foot per second of velocity. Lose a foot per second for 20 seconds and you've been beaten by almost two car lengths. Make these one-foot line mistakes in every corner and you're out of contention.

How close you get to the edges of the road will vary by circumstance. You might not choose to apex as closely to the wall at a street circuit like Long Beach as you might at a benign course like Topeka. But you should be making the *choice* not to get too close, not missing the apex because you *can't* get there.

You can use every inch of road, and then some—we've seen drivers clout curbs and track right out into the dirt at the exits. It *does* make the radius bigger, but radius is not the only part of the equation. If your line is seriously upsetting the car's grip on the road surface, or increasing the likelihood of damaging the suspension, the extra radius is probably not worth it.

Turning Too Early

Let's take a look at Turn 7 again and see what happens when you keep the car on the edge of the road at the corner entry, but initiate the turn too soon.

The entry phase of this corner works just fine with an early turn-in. You're driving into the corner on an arc with a much bigger radius to it. If you're going about the same speed you were going on the last lap, the car would feel better, that is, it wouldn't take as much force to turn the car on a more gentle arc. It may feel good, but at some point it becomes obvious that you're heading off the road. Your sense of self-preservation urges you at this point to turn the steering wheel more to keep the car on the racetrack.

The key question here is whether the car can do it or not. If you're going slowly, not using all the traction that the tires have available, the car has the surplus turning power available for you. You turn more and the car responds, turns on the tighter arc, and stays on the road.

But, if you made this mistake at the limit of the adhesion of the tires with a 1 G-cornering car, and your ideal arc is 195 feet, this early arc would scale to 300 feet. At the car's cornering limit, you could carry 67 m.p.h. into the corner on the 300-foot arc—13 m.p.h. more than if you turned in at the later turn-in point.

The problem is that to stay on the racetrack, you have to negotiate a tight arc late in the turn. The radi-

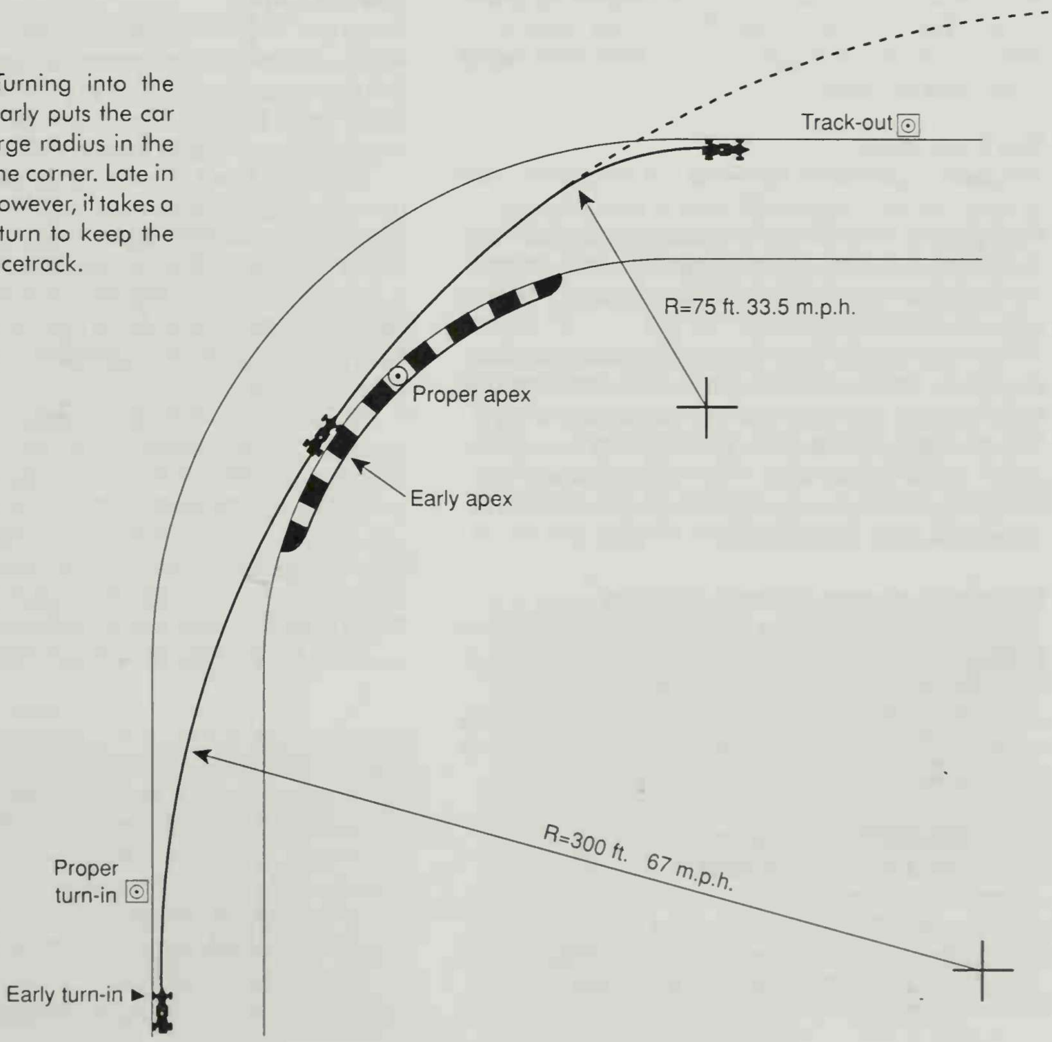


Fig. 2-4. Turning into the corner too early puts the car on a very large radius in the first part of the corner. Late in the corner, however, it takes a tight radius turn to keep the car on the racetrack.

"I've got a great picture at home of the GTO Cougar I used to race. It's a picture of the front wheels around an angled wall at Del Mar where you'd swear the door was touching the wall. It was so close to touching that the reflection off the orange paint is on the wall and you can't see any light through. The car had bulbous fenders and the door went in a little bit between the front and rear wheels and I used the indentation in the door between the fender flares to get closer to the apex."

—Dorsey Schroeder

us of that arc scales to 75 feet: using the cornering equation, you get 33.5 m.p.h. as the maximum speed on this arc. You can't negotiate a 33.5 m.p.h. corner at 67 m.p.h.

If you turn the steering wheel more, the front wheels will point more to the right but since the car is at its cornering limit, it will continue on its way on the 300-foot arc you started back at the beginning of the corner. Sadly, that arc leads off the racing surface. What becomes of you and your racecar is at the mercy of the laws of physics.

The Early Apex

This unfortunate chain of events is known as the "early apex." By turning into the corner too early, you have formed an apex that is significantly earlier than the proper one. The two are naturally tied together: turning in too early will lead to an early apex; an early apex leads to an early exit.

An early apex is by far the most common mistake that gets both novice *and* experienced race drivers off the racetrack. The next time you are at a racetrack look for the areas where no grass has ever or will ever grow—at the corner exits. Why? Because everyone continues to drop wheels, or entire cars, off at the exits because they early apexed.

Primary Symptom of Early Apexing

If you're driving through a corner and feel the need to increase the amount of steering effort past the apex, it is probably because you turned into the corner *too soon*. This is the single most reliable symptom of early apexing; you can use it from your first day in a racecar until the day you hang up the helmet.

Correcting the Early Apex

Recognizing the symptom means you need to do something different the next time you drive up to the corner—namely, turn later. In order to do this you need to firmly establish your reference points *and* develop a sense for where you are on the racetrack.

One other thing you need to do is to examine your perspective or attitude. You *guaranteed* that your car would go off into the boonies because you asked it to do something it was flat incapable of doing: take a 33.5 m.p.h. arc at 67 m.p.h. If you had driven into the corner at 33.5 m.p.h. and late in the turn realized that you needed the tighter arc to stay on the road, you would have gotten away with the mistake. The car would have turned on the 75-foot arc you wanted. You made the same mistake but it cost you less. And the reason it cost you less was not that you didn't screw up, you still early apexed but you had a cushion. You were operating under the limit, using less than 100% of the car's capability, keeping a reserve in hand in case of foul-up.

You can't operate with this reserve forever. A racecar driven at lap-record pace is being used with no reserve. But you're not qualifying now. You're looking for the proper path around the racetrack. You can afford to leave yourself a cushion, especially if it points the way to the proper line and keeps you on the racing surface.

The Late Apex

Using our Turn 7 example, let's look at what happens when you turn into the corner too late. If you turn 20 feet past the geometric turn-in point, the first thing you'll find is that you clip an apex 20 feet further around the curb: a later turn-in will lead to a later apex.

At the track-out, if you held the steering wheel in the same position from turn-in until you're facing down the following straight, you'd have almost 20 feet of racetrack left to the outside at the exit of the corner! From a safety standpoint, that's terrific. You make a mistake and you don't go off the track—you have extra road left. Therein lies the beauty and value of the "late apex."

There are, however, disadvantages. Compute the potential speed you could carry into the corner on an arc that starts 20 feet later than perfect. By scaling it you find that your 20-foot later turn-in requires that you use an arc with a radius of 160 feet to make the apex. Plugging 160 feet into our cornering speed equation yields a maximum speed of 49 m.p.h. That's 5 m.p.h. slower than the geometric line. So the late apex line is a safer but slower path through the corner.

"Absolutely, you need reference points. You can't do it all by feel. You have to find your own marks on the track—that's how you'll be able to go fast consistently. There are lots of places where it's hard to find markers. At a place like Cleveland that's flat and featureless, you have to try even harder to find your spots."

—Danny Sullivan

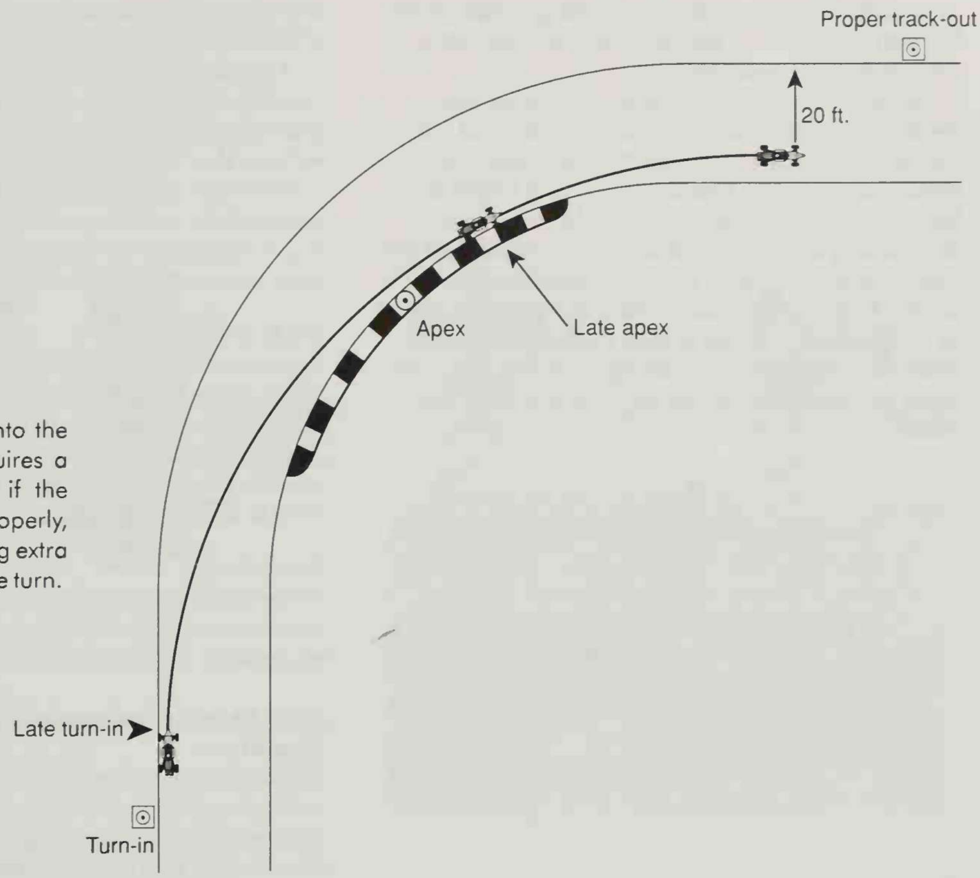


Fig. 2-5. Turning into the corner too late requires a tighter radius than if the corner were done properly, but it results in having extra room at the exit of the turn.

Primary Symptom of Late Apexing

If there is road left at the exit of the corner, you have chosen a turn-in and apex that were *too late.*

Now, this assumes that you're driving at less than the limit. You know by now that any racer would naturally unwind the steering wheel when he or she realized there was 20 feet of road left at the corner exit. As you get closer to the limit, the symptom becomes more like the suspicion that driving away from this corner is too easy. You're able to essentially drive straight out to the edge of the road.

At line-finding speed, the late apex is an advantage since you're looking for reference points and not lap time. You're much better off erring on the side of turning in and apexing too late than too early.

Line-Finding Strategy

Since the late apex is a safer path to take through the corner, it's a very useful tool to use in the effort to find the right line around the racetrack. When you are at a circuit that's new to you, intentionally turn into corners later than you think you should. You'll have a sense (a combination of hand-eye coordination and experience) for when you should initiate a turn.

When you feel you should turn in, overrule your instinct and turn later. Aim for an apex point that's later than you expect it eventually will be, then hold the steering wheel in the same position and see what happens at the exit. If the apex was too late, turn a bit earlier on the next run up to the corner, and aim for an earlier apex. Continue the process lap after lap, adjusting the turn-in and apex until the car is using all the road coming out of the corner. All along you should be picking up reference points for the turn-in, apex, and track-out and working from them, adjusting your line. After many repetitions, you'll develop the knowledge of where the car should be, every inch of the way around the course.

> *"At a new racetrack I look closely at the course and try to find references for turn-ins and apexes and start out, just as you learn in school, later than you might think. You always start later, then back it up earlier and earlier, fine tuning the line."*
>
> *—Brian Till*

Make Time

Those of you who are active racers may say, "Who's got the time? Practice is only a half hour, then we have to qualify." There is always time pressure. Just remember that to get the car on the pole you'll have to be driving the optimum line.

Make time. Almost all tracks have open practice days. If they don't, take your family sedan and enter a car club time trial event. Use the track time to drive around and find the line. Don't feel like it's Mickey Mouse. Guys like Emerson Fittipaldi will drive around a new circuit for hours in a rental car just getting a feel for the place. The line you take at 40 m.p.h. in your street car may not be the exact path you'll take in your racecar, but you'll have reference points to work from and when you finally go racing, the racetrack will feel more like an old shoe than a foreign planet.

> *"When you go to a racetrack for the first time, you just have to go there early and drive around it. I've done everything from driving it in a rental car to riding it on a bicycle or a motorbike. If you don't, you're stuck using the first practice session in the race car, driving slowly, learning it, and getting the line down without doing it at breakneck speed."*
>
> *—Danny Sullivan*

EXIT SPEED

Close to the Edge

Once you're confident of your ability to put the car on the line, you can more safely move up toward the car's cornering limit with less fear that you're making a commitment to the corner that the car can't keep.

In the earlier Turn 7 example, you started out looking for the line by driving through the corner in the 40 m.p.h. range. This is 14 to 17 m.p.h. slower than the speed you can expect to reach at the limit. Being this conservative on speed means that the tires only need to use up 55% of their cornering traction to negotiate the proper line.

To maximize the speed at the corner track-out, you'll eventually have to use everything the car has got. To do this, you have to develop some familiarity with how the car behaves and how you can affect its behavior when it's being driven near its limit using a combination of *throttle* and *steering*.

Neutral

We said originally that as the car went faster and faster it used up more and more of its 100% of available traction, and that there was a specific speed where it would have to use 100% of its hold on the road to stay on its cornering arc. We assumed in this example, for the sake of simplicity, that the front pair of tires and the rear pair of tires on the car reached this 100% level of traction at the same instant.

This *does* happen in the real world, but not often. Usually, one end of the car reaches its limit of traction a bit before the other end and this affects what we refer to as the "cornering balance" of the car.

However, in the case of a car that reaches the limit at both front and rear simultaneously, we call it a neutral handling car: both front and rear tires contribute their maximum traction to the cornering effort.

Understeer

If you were on the skidpad with a car which reached 100% of the tires' cornering capabilities at the front tires before the rears, the front end of the car would slide first, leading you nose-first away from the direction the front wheels were pointing. This cornering state is called "understeer."

Oversteer

At the other end of the spectrum is the case where the rear tires reach the limit first and slide out wider than you intend. This tail-out attitude is called "oversteer."

How Throttle Influences Understeer Or Oversteer

Different cars, owing to their particular design and set-up, have built-in tendencies to either oversteer or understeer at the cornering limit. In most cases, however, the car isn't grossly biased toward one extreme, and you can use the controls of the car to dial up whatever cornering balance is needed. To do this you have to have some effect on the traction of the front or rear pair of tires. It would be nice if you had a dial on the dash to do this, but you don't. The next best thing is to use the control you have down in the footwell of the car, the throttle.

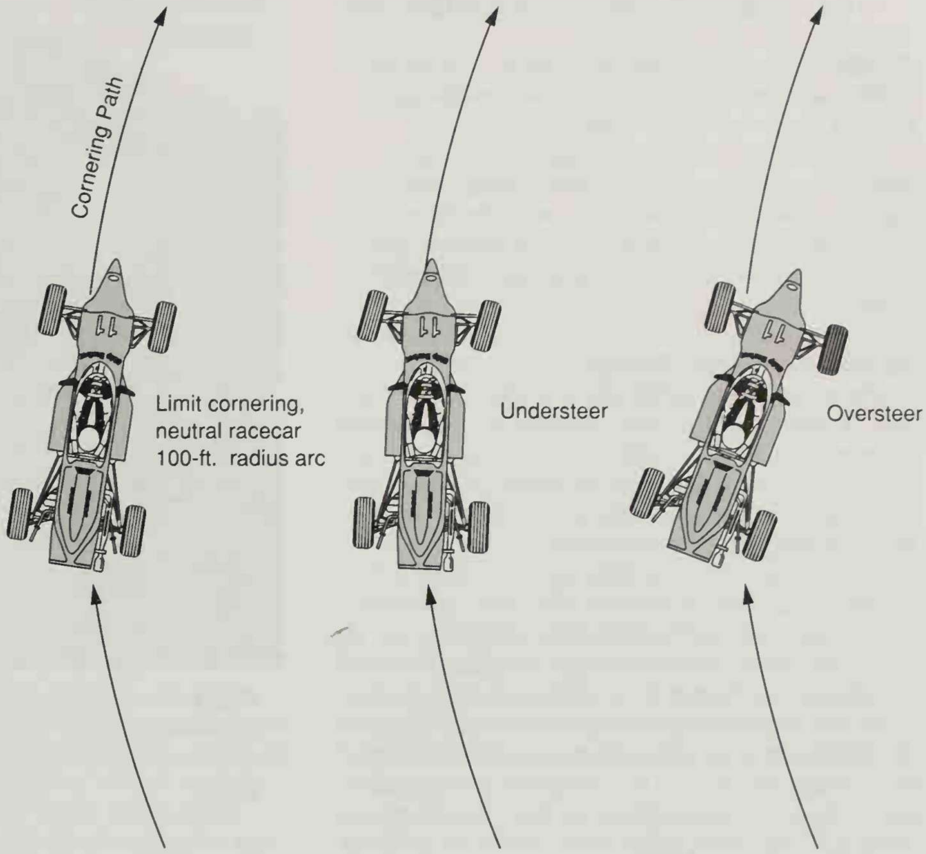
You can load or unload the front or rear tires (change their traction) by applying or backing off the throttle. You can also steal cornering grip from the rear tires (assuming a rear wheel drive car) by applying the throttle and demanding acceleration, or by backing off the throttle and demanding engine-braking traction.

In a rear wheel drive car, applying the throttle creates two opposing tendencies: first, it transfers load onto the rear tires, which helps them grip, but at the same time it uses up part of the traction, which hurts cornering grip. Whichever of the two has the greater effect will determine the result.

For example, if you're driving an underpowered car with sticky rear tires, those tires don't have to surrender much grip to deal with the level of acceleration the motor creates. The increase in grip from the transfer of load, combined with the loss of grip at the front is likely to produce understeer.

If, on the other hand, you're driving a car with skinny rear tires and a King-Kong motor, you can use

Fig. 2-6. When both front and rear tires are contributing proportionately to the cornering effort of the car, its cornering balance is *neutral*. *Understeer* occurs when the front tires reach their limit before the rears. When the rear tires reach their limit before the front tires, *oversteer* occurs.



up a giant percentage of rear tire grip with the same level of squeeze on the throttle, producing oversteer.

To pin it down to a fundamental rule (remember, this is a guideline, not a guarantee):

A gradual increase in throttle will tend to create understeer. An abrupt application of throttle will tend to create oversteer.

Trailing Throttle Oversteer

Now let's look at throttle release. When you release the throttle, relaxing pedal pressure, there is a point where you're using just enough engine power to maintain speed, neither accelerating nor decelerating. The front to rear loading is constant.

If you back off further on the throttle, engine compression slows down the driven wheels and the resulting deceleration shifts load off the rear tires onto the front. Along with the change in loading, the engine compression uses some of the rear tires' grip for braking, further reducing traction at the rear.

If you're cornering near the limit, these changes will cause the rear of the car to slide out onto a bigger arc—into oversteer. This is known as "trailing throttle oversteer," oversteer caused by trailing off the throttle. The more abrupt the throttle lift, the more violent the oversteer. A slight breathe off the throttle would result in a smooth transition to oversteer, while a snap off the throttle can snap the car around faster than you can react to it.

To pin it down to another fundamental rule (remember, this is a guideline, not a guarantee):

Lifting off the throttle while near the cornering limit will create oversteer in direct proportion to the severity of the lift.

Using Throttle to Correct Understeer

As you work towards raising your exit speed out of corners, you'll experiment with squeezing the power on earlier and earlier. As you get closer to the limit, you're likely to experience understeer since your application of power unloads the front end of the car. Behind the wheel, the sensation is one of the car running wider than you intend it to.

The most common reaction is to turn the steering wheel more: after all, past experience has taught that the more you want to turn, the more you turn the steering wheel. But when you are over the front tire's cornering limits, more steering angle is *not* going to make the front end turn more.

The front tires need more download in order to provide more grip. The dilemma is that you don't want to give up pushing on the throttle coming off the corner since it's going to hurt your exit speed. At times like these you have to remember that going off the road hurts exit speed even more.

In the case of developing understeer at the exit of the corner, the right thing to do is to surrender some throttle—often not much—to transfer a little load back to the front tires and get the nose of the car pointed back toward the inside.

Dealing with understeer frequently is done by adjusting the throttle, not the steering wheel. It's probably the easiest car control to develop, since it parallels your instinct to slow down when things aren't going well. Dealing with oversteer typically takes a little more practice.

Steering to Correct Oversteer

On the racetrack it's called oversteer, but most people call it a skid. The street driver is interested in stopping the oversteer before it develops into a loss of control, and doesn't care if the car stops in the process. The street driver is best advised to use only the steering wheel to correct an oversteer slide.

On the racetrack, you certainly don't want to lose control either, but you also don't want to lose very much speed or time. These slightly different goals affect the way you use the steering wheel *and* the throttle to keep oversteer from developing into a spin.

The slide starts somewhere. For some reason—trailing throttle, too much throttle, a missed downshift, whatever—the tail end of the car moves out onto a bigger arc than the arc the front wheels are on. What is the first thing you do?

Back in Driver's Ed., most of us learned to "steer into a skid," and this advice is correct—up to a point. As the rear end comes out, you turn the steering wheel toward the direction the rear end is heading. You're trying to lessen the amount that the front end is being drawn toward the inside of the corner, exaggerating the effect of the rear coming out. By turning the wheel toward the direction the rear is going, you're trying to put the front end of the car on the same arc that the rear is heading for, which will bring the oversteer situation back closer to neutral.

Fig. 2-7. As the rear of the car slides wide, the driver steers the front wheels toward the direction the rear is going. Here the "Correction" is over 270 degrees.

Correction, Pause, Recovery

Turning the steering wheel is the "correction" phase of the slide. If you've corrected enough, the motion of the rear end toward the outside will slow and stop at some point, then begin a swing back toward the inside. This point is called the "pause," and is the signal to begin the "recovery," where you unwind the steering wheel, taking the correction back to zero as the rear of the car comes back from its slide

Note that as the rear end begins its swing back, toward the inside of the arc, it builds up momentum. If you're slow to take out the steering correction that halted the first slide, this momentum will snap the car around in a direction opposite the first slide.

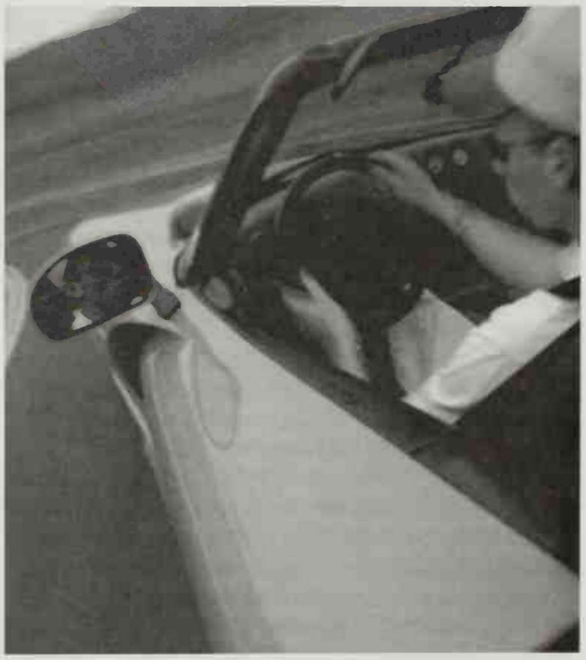


Fig. 2-8. At the height of the slide, the rear end pauses for an instant before coming back the other way. This "Pause" is the cue to begin the "Recovery'— bringing the steering wheel back to straight, quickly.

Every tail-out slide should be dealt with by making a correction, then using the pause as a cue for beginning the recovery. While these phases primarily define what you should do with the steering wheel, the throttle can come into play.

Combining Steering and Throttle for Control

Exactly how you use the throttle depends on what initially caused the oversteer. If the tail came out because you abruptly lifted off the throttle in the middle of the corner, causing trailing throttle oversteer, you can contribute to restoring traction to the rear of the car by applying some throttle.

If the oversteer began because you got greedy on the throttle and created power oversteer, you can help solve the problem by breathing back on the throttle, which should restore some cornering traction to the rear of the car.

Using the throttle correctly takes a tender touch. A majority of both beginning and experienced drivers over-use the throttle in these situations. The most common misuse is full power to solve any situation. You need to remember that small changes in throttle can have a huge effect on the car, especially when it's dancing through a corner on the limit of adhesion.

So far the concentration has been on car control in the part of the corner where you combine turning and accelerating, the area where beginning drivers should first experiment with the cornering limits of the car.

You'll soon see that the car is also driven at its combined limit of braking and cornering on the way from the turn-in point of the corner to the throttle application point. You have to use car control to keep the car balanced at its limit in this part of the racetrack as well, but you'll find that it's harder to creep up on this limit than it is to flirt with the limit of combined acceleration and cornering. You'll also find that there is less overall lap time to be gained by being right on the limit at corner entries than there is to be gained being on the limit coming out of corners.

> *"Using both your hands and the throttle is something that is totally feel. The first thing is opposite lock—turn into the slide. Then you balance the throttle to help get the car as straight as you can, but it's like dancing on the throttle—little changes either more or less according to what the car needs."*
>
> *—Danny Sullivan*

BRAKING AND ENTERING

The final step in lowering lap time is developing braking skill. This aspect of driving skill is often referred to as "braking and entering."

Lose Speed Quickly

At corner entry there are two fundamental goals. The first is to minimize the time it takes to go from the brake application point to the throttle application point. The assumption here is that spending less time on the brakes means more time on the throttle, which *has* to be faster. This approach would mean using all of the braking power available so that you can lose the speed required in the shortest amount of time.

Lose the Right Amount of Speed

Just as important as the rate of speed loss is the total amount of speed you choose to lose. If you slow too much by either braking too early or too hard or too long, this speed is lost not only on the corner entry but on the acceleration zone following the corner.

Not slowing enough can put the car off the racing surface. Arriving in the corner too fast will, at the very least, delay the beginning of the acceleration away from the corner as you struggle for control on the way to the apex.

Braking Fundamentals

It's helpful to know some of the forces at work when you use the braking system, and what will be required of you to approach the limit under braking.

As with cornering and accelerating, it is tires' grip against the road surface that provides you with the force to slow the car. Put a better set of tires on the car—tires that have a greater capability for grip—and the car's braking performance will improve.

Tires create traction in direct proportion to how hard they are pushing down on the surface of the road. If, for example, a tire is pushing down on the road surface with 500 lbs. of force, a typical race tire will provide you with 500 lbs. of maximum resistance against the road surface.

You can use this potential 500-lb. force however you choose. You can use it to corner the car or to accelerate it. As you've seen earlier, you can use a combination of accelerating and cornering, or braking and cornering. Braking in a straight line, you can use all of this 500-lb. force exclusively for braking.

Pressing on the brake pedal creates hydraulic pressure in the brake lines and calipers, pressing the brake pads against the brake rotors which are spinning around at the speed of the wheel and tire. The brake system resists this rotation. The force exerted by the brake system is in the opposite direction from the rotation of the tire, illustrated by the clockwise arrow at

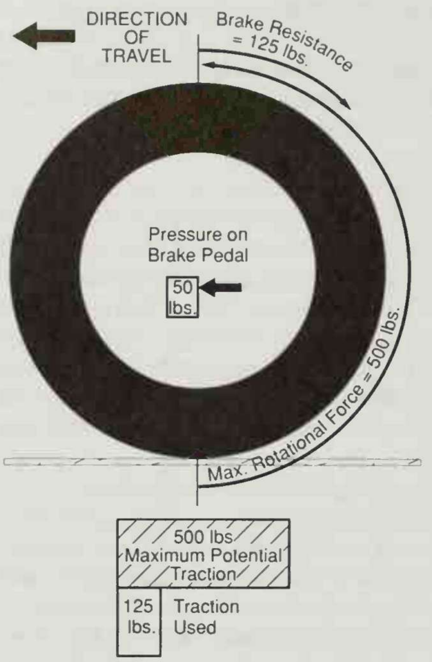


Fig. 2-9. With light brake pedal pressure the resistance to the tire's rotation is less than the tractive force that the tire could exert at the limit of its grip on the road.

the top of Fig. 2-9. You control the amount of this resistance by the pressure exerted on the brake pedal.

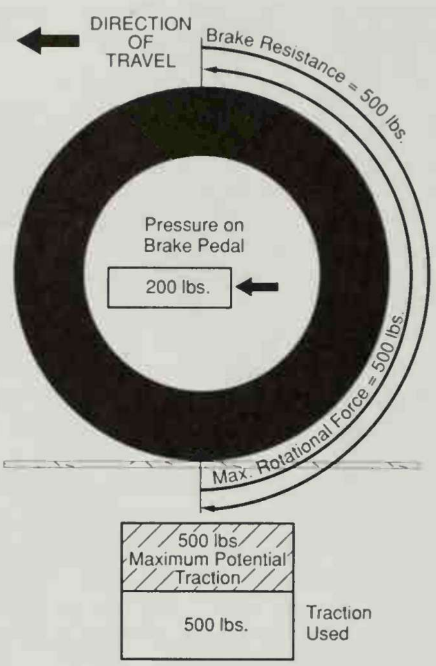
If you push lightly, with 50 lbs. of pressure on the brake pedal, it would only take 125 lbs. of the potential 500 lbs. of force to turn the tire counterclockwise, leaving 375 lbs. of potential traction in reserve.Doubling the pedal pressure to 100 lbs. forces the tire to use 250 lbs. of force to rotate the wheel.

As the brake pedal pressure goes up, the force countering the rotation of the tire goes up proportionately (see Fig. 2-10). You'll notice too that the force on the pedal is multiplied by the design of the hydraulic system. 100 lbs. of force on the pedal, for example, creates 250 lbs. of force at the tire.

As you push harder on the brake pedal, more and more of the potential 500 lbs. of traction has to be used to counter the resistance to rotation. You can see from the chart that there is a specific pedal pressure at which it will take *all* of the tire's grip to keep the tire turning. If your aim is to slow down in the shortest possible time and distance, this is the place you want to be. At around 200 lbs. of pressure on the brake pedal, you're using all 500 lbs. of potential force to slow the car.

Threshold Braking

The goal is to stay at this maximum traction point. This occurs when the tire is revolving at a rate 15%



Brake Pedal Pressure (in lbs.)	Brake System Force (in lbs.)
50	125
100	250
150	375
200	500
220	550

Fig. 2-10. Higher pedal pressures use proportionately more of the potential grip. Here, the brake resistance equals the maximum rotational force, signifying threshold braking.

slower than it would be if it were freely rolling over the surface. This is a very tender balance point. Any less pressure on the brake pedal, and less than the full 500 lbs. of braking traction would be used, making the speed loss take more time than necessary. At a particular level of pressure on the brake pedal you're right there at the peak. We call it the *threshold*.

Lockup

If you go beyond the maximum and increase the pedal pressure past 200 lbs., the resistance to the tire's rotation is greater than the force acting to revolve the tire and the tire stops turning. When "tire lockup" happens, things go from bad to worse.

When the tire stops rotating, it typically loses 30% or more of its grip. Your original potential of 500 lbs. of force drops to 350 lbs. It takes longer to slow down, not to mention the damage you're doing to the tire by grinding the rubber off the spot in contact with the pavement.

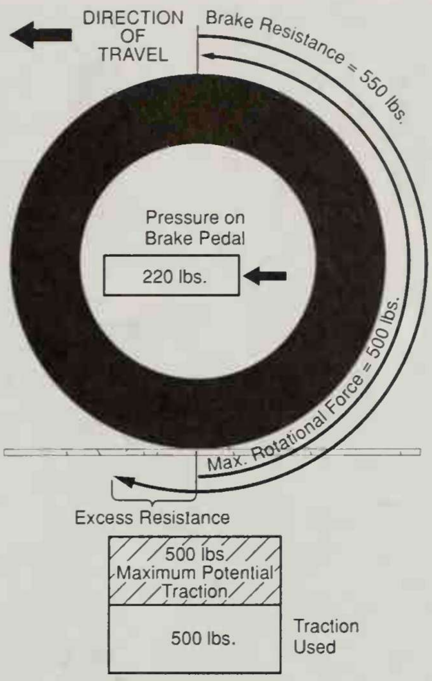


Fig. 2-11. When the resistance to the rotation is greater than the maximum tractive force of the tire, the wheel stops rotating.

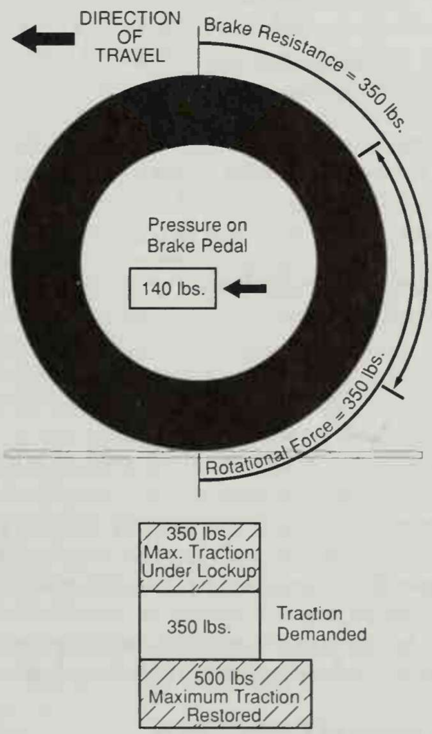


Fig. 2-12. When the tire stops rotating, the tractive force falls by some 30%. By relaxing the brake pedal pressure, you eliminate lockup and restore full braking traction.

Brake Modulation

In order to get the tire rotating again, you need to reduce pressure in the braking system enough so that this 350 lbs. of drag against the contact patch will start the tire rotating again. That is not to say, "take your foot off the brake pedal" (the most common beginner mistake). Pressure on the brake pedal needs to be reduced but not eliminated. Once the tire begins to roll again, its full traction is restored and you can increase pressure again in search of that level that puts the tire back to its threshold, yielding the maximum braking grip available.

From a theoretical standpoint, threshold braking is matching the appropriate brake pedal pressure to the maximum grip of the tire. In the real world things get complicated since you're dealing with not one, but two pairs of tires on opposite ends of the car.

Brake Bias

It's rare to have all four tires doing an equal share of the braking. When cars change speed and direction, inertia transfers load from front to rear and side to side. These loading changes alter each tire's traction.

Take a look at Fig. 2-13, a diagram of one of our Formula cars as it sits motionless in the pit lane. Its 1000-lb. mass is distributed as follows: 600 lbs. on the rear set of tires and 400 lbs. on the front pair. This is its static weight distribution which, once the car moves, is almost never seen again.

As the car accelerates, inertia (which wants the car to stay at rest and resists attempts to change its velocity) transfers load off the front tires onto the rear tires. The amount transferred depends upon the center of gravity of the car, its wheelbase, and the level of acceleration. When a car decelerates, the load is transferred off the rear tires onto the fronts. The greater the level of deceleration, the greater the load transfer off the rear tires to the fronts.

Under maximum deceleration, one of our Formula cars may transfer 250 lbs. of load onto the front tires, making the download 650 lbs. on the fronts and 350 lbs. on the rears; under hard braking 65% of the traction is at the front pair of tires while only 35% is at the rears. It would seem appropriate, then, that the front brakes exert 65% of the total braking effort while the rears take care of the remaining 35%. This difference in the proportion of braking effort afforded the front vs. the rear is called "brake bias."

Most street cars have this proportioning done by the designers of the car and the bias is figured on a particular loading and target range of deceleration. For racecars which are pursuing the maximum available performance, this kind of "ballpark" brake bias isn't good enough.

All real racecars will have a brake bias adjustment on the car (see Fig. 2-14). Some are easily driver-adjustable, while others require a stop at the pits and a mechanic's help. The aim is to get both ends of the car

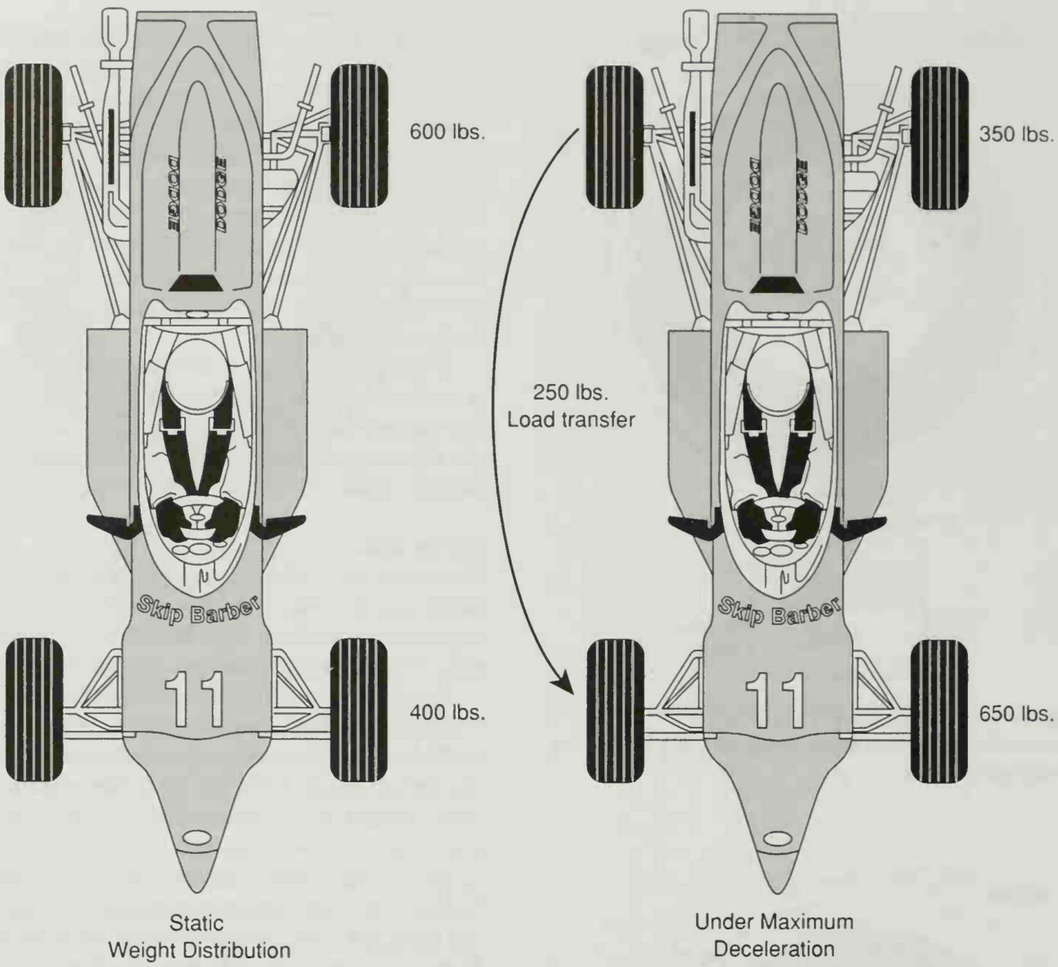


Fig. 2-13. Sitting stationary in the pit lane, the static loading on the tires is different than it will be when you put the car into motion. Under deceleration the loading of the front versus the rear tires changes, transferring load from the rear to the front.

doing their fair share of the braking. If the front tires have 65% of the traction yet they have 75% of the brake effort, the rear tires aren't doing their full share and the front tires will tend to lock prematurely. The same holds true for the rears.

Of the two, front bias is easier to deal with than rear bias. If you inadvertently lock a front wheel, the car at least will continue to go straight. The car won't turn because the locked tire can't deliver a lateral force to change the car's direction and the rears, with the drag of both brakes and engine drag, will act like the feathers on an arrow, keeping the rear wheels obediently behind the fronts.

Lock the rears, however, and things like road crown or crosswinds or the slightest steering wiggle will get the rear sliding. It's not fun, especially at the end of a straightaway where you'd like to be thinking about getting the car slowed for the rapidly approaching corner, not catching an oversteer slide.

How Much Different from the Street?

You know that braking on the racetrack is going to be different than normal street driving because you're going to use a lot more force than you're accustomed to. Other than the level of force, there is nothing new here in the basic process.

Be forewarned that you will never be able to come up with a corner entry routine that covers all cases. In some instances, you arrive at the end of the straight at 180 m.p.h. and need to slow for a 25 m.p.h. corner. In some cases the approach is at 70 m.p.h. and the corner requires 68 m.p.h. It should be no surprise that different techniques are required for these two extreme cases, and for the others that lie in between.

Components of Corner Entry

Even with the wide variety of corner entries in racing, there are basic components which are common to most speed-loss situations. We can break the whole process down into four major blocks which might all

Fig. 2-14. Brake bias adjusters allow the driver to alter the brake bias of the car while it's underway.

get used. But, as you'll see, there are corner entry situations where you will choose to use just one of the four blocks. For now, a simple explanation will suffice. Later, we will look closely at each of these blocks and see what types of skills you need to develop in order to excel in this aspect of the racing game.

Block 1: The Throttle-Brake Transition. Taking your foot off the throttle and putting it on the brake pedal can be done in very many different ways. Here, you're covering the range of techniques from a slow, cautious, lift off the throttle and onto the brakes, to the lightning-quick transition from full throttle to hundreds of pounds of brake pedal pressure.

Block 2: Straight-Line Deceleration. When the car is going straight it has no other demands on the tires than braking. You can decelerate at the car's maximum rate in this segment of corner entry. There are frequently cases when you will want to exercise the option to brake at less than the "threshold" level.

Block 3: Brake-Turn. Here you can both decelerate and change direction. In brake-turning, you can use a constant level of pressure on the brake pedal, yielding a constant level of cornering force, or you can reduce the braking pressure over time, creating increasing cornering force as the car approaches the throttle application point.

> *"I check the brake bias just about every time I get in a race car. Stuff always changes. Some cars are wildly sensitive to brake bias and some aren't, but you have to check. It also depends on the racetrack as well. If I go to a place like Road America where there are four or more hard straight-line braking zones, I'm going to try more rear brake bias in the car to absolutely maximize the car's straight-line braking potential. At a place like Lime Rock where there is no straight-line braking zone and you are braking and turning a lot, I'll run more front bias so that I can carry more brake power into the corner with more confidence."*
>
> *—Jeremy Dale*

Block 4: Brake-Throttle Transition. At the very end of deceleration you leave the brake pedal and pick up the throttle to accelerate away from the corner. You can do this quickly or slowly, with or without a pause between the last bit of braking and the first bit of throttle.

THE ANALYTICAL RACER

In this chapter, we have introduced some of the most basic information a good race driver should understand in order to do well. As you can see, there is a lot to it, and performing well has a lot less to do with instinct than with planning.

A good driver knows what needs to be done at every point on the racetrack and takes the time to develop the skill needed to make the proper moves. There is a plan: brake there; shift there; turn in at that spot; take a specific apex; apply the power just so; clip the edge of the road at the exit.

Early in a career many of these decisions take a good deal of conscious thought. As you get more experience and develop the physical skills necessary to drive a car fast and smoothly, less conscious thought is required for these decisions, and more can be applied to advanced subjects that will affect your level of success.